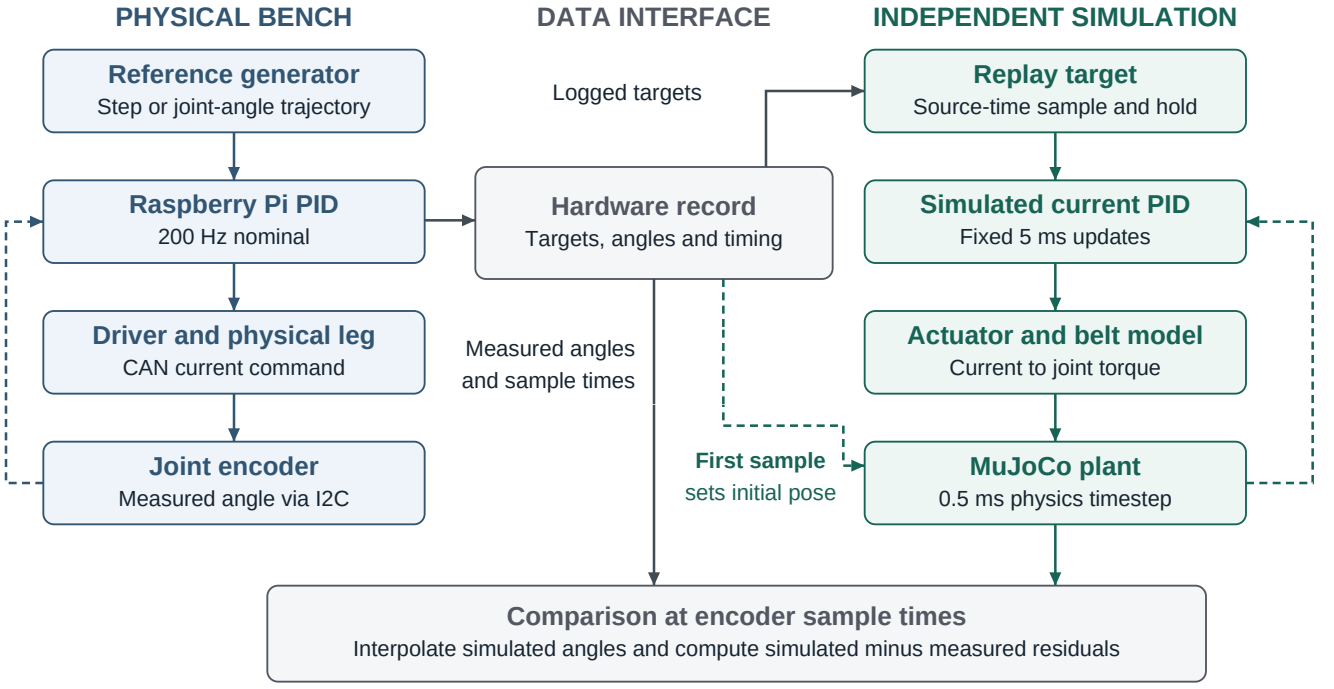
PHYSICAL BENCH
DATA INTERFACE
INDEPENDENT SIMULATION
Reference generator
Step or joint-angle trajectory
Raspberry Pi PID
200 Hz nominal
Driver and physical leg
CAN current command
Joint encoder
Measured angle via I2C
Logged targets
Hardware record
Targets, angles and timing
Measured angles
and sample times
First sample
sets initial pose
Replay target
Source-time sample and hold
Simulated current PID
Fixed 5 ms updates
Actuator and belt model
Current to joint torque
MuJoCo plant
0.5 ms physics timestep
Comparison at encoder sample times
Interpolate simulated angles and compute simulated minus measured residuals